Year 9
Mathematics

Unit 3: Sequences and Algebra

3.5 Solving Quadratic Equations

Solving Quadratic Equations

STARTER · 5 MIN

Date: _____ **Title: Solving Quadratic Equations**

LEARNING OBJECTIVES

- Solve a quadratic equation by factorising
- Solve equations with negatives and $x^2 - a^2 = 0$
- Form and solve quadratic equations

STARTER *Answer in your book. No calculators.*

1. Factorise $x^2 + 5x + 6$
2. Solve $x + 3 = 0$
3. Solve $2x - 8 = 0$
4. Work out 9^2 and $(-9)^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $(x + 2)(x + 3)$

2. $x = -3$

3. $x = 4$

4. 81 and 81

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve a quadratic equation by factorising	☐
SILVER	I can solve equations with negatives and $x^2 - a^2 = 0$	☐
GOLD	I can form and solve quadratic equations	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Solving Quadratic Equations

INTRODUCTION · 3 MIN

If two numbers multiply to give 0, one of them must be 0.

Factorise

Make one side 0, then factorise.

$x^2 + 10x + 21 = 0 \rightarrow (x + 3)(x + 7) = 0$

Solve

Set each bracket equal to 0.

$x + 3 = 0$ or $x + 7 = 0$: $x = -3$ or -7

$x^2 - a^2 = 0$

Two answers: + and −.

$x^2 - 81 = 0$: $x = 9$ or -9

Rearrange

Move every term to one side first.

$x^2 + 6x = 16 \rightarrow x^2 + 6x - 16 = 0$

Examples 1 and 2 — Solving by Factorising

EXAMPLES · 5 MIN

BRONZE *Watch and copy into your book*

Hint if A × B = 0, then A = 0 or B = 0

Example 1

Solve

$(x + 3)(x - 5) = 0$

Example 2

Solve

$x^2 + 10x + 21 = 0$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

One bracket must be 0

$x + 3 = 0$ or $x - 5 = 0$

$x = -3$ or $x = 5$

Tip: the signs change: +3 gives −3.

Example 2

Factorise first

$(x + 3)(x + 7) = 0$

$x + 3 = 0$ or $x + 7 = 0$

$x = -3$ or $x = -7$

Tip: check: $9 - 30 + 21 = 0$

Task 1

PRACTICE · 9 MIN

BRONZE *Solve*

Hint if A × B = 0, then A = 0 or B = 0

1. $(x + 2)(x - 4) = 0$

2. $(x - 1)(x - 6) = 0$

3. $x^2 + 5x + 6 = 0$

4. $x^2 + 7x + 12 = 0$

5. $x^2 + 8x + 15 = 0$

★ $x^2 + 9x + 14 = 0$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $x = -2$ or 4

2. $x = 1$ or 6

3. $x = -2$ or -3

4. $x = -3$ or -4

5. $x = -3$ or -5

★ $x = -2$ or -7

Examples 3 and 4 — Negatives and $x^2 - a^2 = 0$

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $x^2 - a^2 = (x + a)(x - a)$ −c: different signs

Example 3

Solve

$x^2 - x - 12 = 0$

Example 4

Solve

$x^2 - 81 = 0$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

−12: different signs

$(x - 4)(x + 3) = 0$

$x = 4$ or $x = -3$

Tip: $-4 \times 3 = -12$, $-4 + 3 = -1$

Example 4

Difference of two squares

$(x + 9)(x - 9) = 0$

$x = -9$ or $x = 9$

Tip: or $x^2 = 81$, so $x = \pm 9$

Task 2

PRACTICE · 9 MIN

SILVER *Solve*

Hint $x^2 - a^2 = (x + a)(x - a)$ −c: different signs

1 $x^2 - 7x + 10 = 0$

2 $x^2 + 2x - 8 = 0$

3 $x^2 - 3x - 10 = 0$

4 $x^2 - 25 = 0$

5 $x^2 - 144 = 0$

★ $x^2 - 2x - 35 = 0$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x = 2$ or 5

2. $x = 2$ or -4

3. $x = 5$ or -2

4. $x = 5$ or -5

5. $x = 12$ or -12

★ $x = 7$ or -5

Examples 5 and 6 — Forming and Solving

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint rearrange to … = 0 first a length cannot be negative

Example 5

(3 marks)

Solve

$x^2 + 6x = 16$

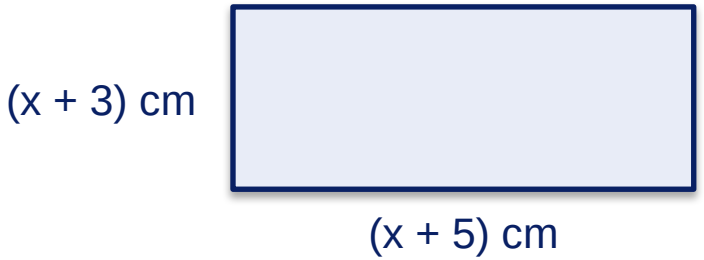
Example 6

(5 marks)

The area is 35 cm².

(a) Show $x^2 + 8x - 20 = 0$

(b) Find the length and width.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Make one side 0

$x^2 + 6x - 16 = 0$

$(x + 8)(x - 2) = 0$

$x = -8$ or $x = 2$

Tip: subtract 16 from both sides.

Example 6

Area = length × width

(a) $x^2 + 8x + 15 = 35$

$x^2 + 8x - 20 = 0$

(b) $(x + 10)(x - 2) = 0$, $x = 2$

length 7 cm, width 5 cm

Tip: $x = -10$ makes the lengths negative.

Task 3

PRACTICE · 9 MIN

GOLD *Solve, or answer the question*

Hint rearrange to … = 0 first a length cannot be negative

1. $x^2 + 3x = 10$

2. $x^2 = 5x + 14$

3. $x^2 - 2x = 24$

4. Rectangle $(x + 4)$ by $(x + 1)$, area 28 cm^2. Find its sides.

5. Square, side $(x + 2)$, area 49 cm^2. Find x.

★ Parallelogram: base $(x + 6)$, height $(x + 1)$, area 36 cm^2. Find the base and height.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. x = 2 or −5
2. x = 7 or −2
3. x = 6 or −4
4. x = 3: 7 cm by 4 cm
5. x = 5 (side 7 cm)

★ x = 3: base 9 cm, height 4 cm

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $(x - 3)(x + 5) = 0$

SILVER Solve $x^2 - 49 = 0$

GOLD Solve $x^2 + x = 12$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **x = 3 or −5**

SILVER **x = 7 or −7**

GOLD **x = 3 or −4**

Exam Practice — Solving by Factorising

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint if A × B = 0, then A = 0 or B = 0

Question 1

(2 marks)

Solve $(x - 2)(x + 9) = 0$

.....

Question 2

(3 marks)

Solve $x^2 + 9x + 20 = 0$

.....

Exam Practice — Negatives and Squares

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $x^2 - a^2 = (x + a)(x - a)$ −c: different signs

Question 3

(3 marks)

Solve $x^2 - 64 = 0$

.....

Question 4

(3 marks)

Solve $x^2 - 4x - 21 = 0$

.....

Exam Practice — Forming and Solving

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint rearrange to … = 0 first
a length cannot be negative

Question 5

(3 marks)

Solve $x^2 + 2x = 15$

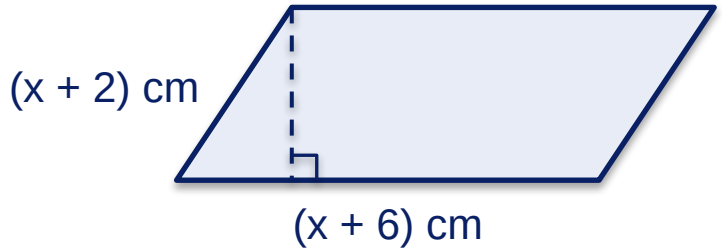
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Question 6

(6 marks)

The parallelogram has area 32 cm^2.

Diagram NOT accurately drawn



(a) Show that $x^2 + 8x - 20 = 0$ (3)

(b) Find the base and the height. (3)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$x - 2 = 0$ or $x + 9 = 0$ — M1

$x = 2$ or $x = -9$ — A1

Tip: each bracket can be 0.

Question 2

$(x + 4)(x + 5) = 0$ — M1

$x = -4$ or $x = -5$ — A2

Tip: A1 for one correct answer.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$(x + 8)(x - 8) = 0$ or $x^2 = 64$ — M1

$x = 8$ or $x = -8$ — A2

Tip: do not forget the negative answer.

Question 4

$(x - 7)(x + 3) = 0$ — M1

$x = 7$ or $x = -3$ — A2

Tip: $-7 \times 3 = -21$, $-7 + 3 = -4$

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x^2 + 2x - 15 = 0$ — M1

$(x + 5)(x - 3) = 0$ — M1

$x = 3$ or $x = -5$ — A1

Tip: make one side 0 first.

Question 6

(a) $(x + 6)(x + 2) = 32$ — M1

$x^2 + 8x + 12 = 32$ — M1

$x^2 + 8x - 20 = 0$ — A1

(b) $(x + 10)(x - 2) = 0$, $x = 2$ — M2

base 8 cm, height 4 cm — A1

Tip: a length cannot be negative.